

Bypassing Automated Audit Verification: Fraud Vector Mechanisms and the Need for General-Purpose AI in Fraud Detection

Abstract

This document explores mechanisms by which fraud vectors in e-commerce transactions are designed to evade automated audit verification. Drawing from analyses of transaction data (e.g., net positives from withheld refunds and repeated debits) and industry examples, it highlights how these vectors exploit flaws in traditional audit software, such as hard-coded patterns that fail to adapt to evolving fraud. The analysis underscores the limitations of rule-based systems and advocates for general-purpose AI, which offers dynamic, adaptive detection. APA-style citations reference empirical sources on audit flaws, fraud mechanisms, and AI advancements.

Introduction

Automated audit and fraud detection software plays a critical role in verifying financial statements, yet its reliance on hard-coded patterns creates exploitable flaws. In e-commerce, fraud vectors like OTP bypasses (+1,126,969.27 INR net) and merchant misclassification (-5,560 INR net) are crafted to mimic legitimate activity, bypassing verification. As evidenced by data patterns (e.g., small probes escalating to 32 lakh INR overcharges), these schemes reveal systemic weaknesses in traditional software, necessitating general-purpose AI for comprehensive detection (West & Bhattacharya, 2022). This document examines bypassing mechanisms through specific fraud vectors, software flaws, and AI's role.

Mechanisms for Bypassing Automated Audit Verification

Fraud vectors are deliberately engineered to pass automated audits by simulating normalcy and exploiting software limitations. Traditional systems use rule-based algorithms (e.g., threshold checks for anomalies), which fraudsters evade through subtle variations. Below, key fraud vectors are detailed with their definitions, overviews, and mechanisms for evasion, drawing from observed patterns in e-commerce data.

1. Missing Identifier Fraud Definition: Missing identifier fraud involves transactions where

payment identifiers (e.g., transaction IDs or authorization codes) are absent or omitted from records, making it difficult to trace or verify payments. Overview: This vector is prevalent in e-commerce platforms where incomplete records allow for unverified or fabricated entries. In analyzed data, 100% of records in certain subsets (e.g., OTP bypass and withheld refunds) lacked identifiers, leading to net discrepancies of hundreds of thousands of INR. It enables revenue inflation by recording phantom payments without audit trails. Mechanism: Identifiers are omitted during processing, allowing misallocation of batch payments to low-value or cancelled orders. For example, repeated amounts without IDs (e.g., 12,467 INR x3 on 2021-07-29) create positives (+37,401 INR) without traceability, exploiting software's data limitations and violating payment standards that require unique IDs for verification (PMC, 2023).

2. **Merchant Misclassification Fraud Definition:** Merchant misclassification fraud occurs when transactions are incorrectly categorized under Merchant Category Codes (MCCs), leading to improper fees, refunds, or compliance issues. Overview: In the data, only two records showed negatives (-5,560 INR) for cancelled orders with zero payments, suggesting misclassification to justify non-refunds. This vector causes merchants to overpay fees or platforms to withhold funds, as seen in industry cases of misclassifying cards leading to penalties (CMSI, 2023). Mechanism: Wrong MCCs are assigned (e.g., high-risk to low-risk orders), blocking refunds or altering fees. Missing payment types in 100% of records enable this, resulting in unprocessed refunds for positive net totals (e.g., 3,560 INR on 2024-08-01), bypassing hard-coded categorization checks (Fraud.net, 2024).
3. **Cash Advances Fraud Definition:** Cash advances fraud involves unauthorized or misrecorded cash withdrawals against credit limits, often converted to EMI with hidden charges. Overview: The data showed net positives (+1,144,278.65 INR) from repeated EMI debits without OTP, retaining funds on cancelled orders. This exploits high-interest advances, leading to overcharges (Wikipedia, n.d.). Mechanism: OTP is bypassed for EMI conversions, debiting multiples (e.g., 92,140 INR x2 on 2022-02-25) without verification, inflating revenue through unrefunded advances and evading threshold-based anomaly detection (India TV News, 2022).
4. **Payment Differs from Invoice Fraud Definition:** This fraud occurs when recorded

payments do not match invoice amounts, leading to over- or under-billing. Overview: Analysis revealed net positives (+2,014,567.42 INR), with payments exceeding invoices on cancelled orders. This creates inflated revenue, as in scams tricking payments for undelivered goods (Precoro, 2024). Mechanism: Records are manipulated to show excess payments (e.g., +105,090 INR x2 on 2022-02-25 for zero net), using paired entries to balance while retaining net gain, fooling aggregate-level audits through batch and offset tactics (Albrecht et al., 2015).

5. **Multiple Debits Fraud Definition:** Multiple debits fraud involves repeated unauthorized charges from the same instrument or amount, often in short bursts. Overview: Data showed net positives (+1,252,239.88 INR) from repeated debits (e.g., 1,197 INR x4 on 2023-06-01), leading to overcharges without corresponding services. Mechanism: The same identifier is used for multiples (e.g., Visa 2431.0 x12 in 2023), debiting bursts to extract funds, with negatives offsetting detection and spacing to avoid burst activity thresholds (Hilal et al., 2022).
6. **EMI Charges Fraud Definition:** EMI charges fraud entails unauthorized or hidden fees on installment payments, often without verification. Overview: Net positives (+1,144,278.65 INR) from unrefunded EMI on cancelled orders, exploiting no-OTP setups for recurring overcharges (India TV News, 2022). Mechanism: OTP is bypassed for EMI debits (e.g., 18,416.96 INR x5 on 2024-04-05), retaining funds post-cancellation without refunds, mimicking legitimate recurring payments to pass pattern-matching (Ngai et al., 2011).
7. **Duplicate Merchant Transactions Fraud Definition:** Duplicate merchant fraud involves repeated transactions to the same or similar merchants, often for laundering or overcharging. Overview: Bursts within 24 hours led to net positives (+930,810.86 INR), with duplicates inflating totals. Mechanism: Multiples are processed with the same type/amount (e.g., 7,915 INR x5 on 2023-02-10 Visa), using offsets to conceal and aligning with statistical norms like bell-shaped distributions (Phua et al., 2010).
8. **OTP Bypass Fraud Definition:** OTP bypass fraud skips one-time password verification

for transactions, allowing unauthorized access. Overview: Net positives (+1,126,969.27 INR) from unverified debits, exploiting no-OTP receipt. Mechanism: System overrides for authentications (e.g., multiples on Visa without OTP) retain funds on cancelled orders, evading verification-focused rules (ReSecurity, 2024).

9. Repeated Amounts Transactions Fraud Definition: Repeated amounts fraud uses identical payment values across transactions to misallocate or duplicate charges. Overview: Net positives (+594,448.60 INR) from duplicates (e.g., 6,377.79 INR x4 on 2025-03-11). Mechanism: Amounts are batched identically (e.g., 18,416.96 INR x5), splitting to inflate positives without unique debits, simulating random errors to pass thresholds (Ngai et al., 2011).
10. Standing Instructions Fraud Definition: Standing instructions fraud manipulates recurring payment mandates, often without records or verification. Overview: Empty files concealed setups, implying unrecorded EMI debits without OTP (GoCardless, n.d.). Mechanism: Mandate records are omitted, processing bursts without OTP and retaining funds post-cancellation, exploiting unstructured data handling flaws.
11. Withheld Refunds Fraud Definition: Withheld refunds fraud retains payments for unfulfilled orders without issuing reimbursements. Overview: Net positives (+952,761.54 INR) from zero net totals but positive payments on cancelled orders. Mechanism: Payments are recorded without refunds (e.g., 169,850 INR on 2022-07-18 cancelled), using repeats to conceal and gradual escalation to avoid spikes (DataDome, 2024).
12. Zero Value Payments Fraud Definition: Zero value payments fraud uses \$0 transactions to test card validity before larger fraud. Overview: Net negatives (-346,008.81 INR) from zero payments on positive nets, often cancelled. Mechanism: Zero amounts are authorized (e.g., multiples on 2025-03-24 Visa), testing before cancelling without refunds, starting with minor probes to evade detection (Stripe, 2024).

These designs exploit software's inability to detect adaptive fraud, as rule-based systems rely on predefined indicators that can be circumvented (Hila et al., 2022).

Flaws in Automated Audit Software: Hard-Coded Patterns

Traditional software relies on static, hard-coded rules (e.g., if-then thresholds for

anomalies), which fail against adaptive fraud.

Key Flaws

- **Inability to Adapt:** Hard-coded patterns miss evolving schemes, like gradual escalations from small probes to 32 lakh INR, producing false negatives (Journal of Accountancy, 2024).
- **High False Positives:** Overly rigid rules flag legitimate activity, reducing trust and allowing real fraud to slip through (ScienceDirect, 2022).
- **Data Limitations:** Software struggles with unstructured data (e.g., missing identifiers in 100% of records), failing to detect concealment. These flaws, as in rule-based systems detecting only 60-70% of fraud, highlight rigidity vs. dynamic threats (ResearchGate, 2024).

The Need for General-Purpose AI in Fraud Detection

General-purpose AI (e.g., models like Grok) addresses these flaws by learning dynamically unlike hard-coded software.

Advantages of AI

- **Adaptive Learning:** AI detects novel patterns (e.g., bell-distributed errors), improving accuracy by 15-30% over traditional methods (ResearchGate, 2024b).
- **Real-Time Analysis:** Processes vast data for anomalies, reducing false positives by 40% (MindBridge, 2024).
- **Holistic Detection:** Integrates unstructured data (e.g., missing IDs, bursts), outperforming rules in e-commerce fraud (VLinkInfo, 2025). AI's superiority is evident in cases like +594,448.60 INR net from repeated amounts, where hard-coded software misses gradual builds, but AI identifies trajectories (FraudFights, 2023).

Conclusion

Bypassing audits through hard-coded flaws and evasive fraud vectors sustains fraud in e-

commerce. General-purpose AI offers a solution by providing flexible, comprehensive detection. Customers detecting via personal probes (e.g., 1.5 crore INR losses) emphasize the urgency for AI adoption.

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